

Statistical Characterisation of a Large-Scale Software Bug Repository

Exploratory analysis of 196,633 Mozilla Bugzilla defect reports underpinning an ML-based bug triage and resolution recommendation system

Dataset: BugsRepo (Zenodo 15004067, CC-BY 4.0) **Source:** bugzilla.mozilla.org REST API **Observation window:** 2021-01-27 → 2024-11-15 (1,387 days) **Unit of analysis:** one bug report

RECORDS ANALYSED	DUPLICATE ROWS REMOVED	PRODUCTS / COMPONENTS	SEVERITY LABELLED	WITH RESOLUTION TIME	MARKED DUPLICATE
196,633	28,563	85 / 725	79,842	153,748	18,796
from 225,196 raw rows	12.68% of raw file	categorical label space	40.6% on the S1–S4 scale	creation → last resolved	non-null dupe_of

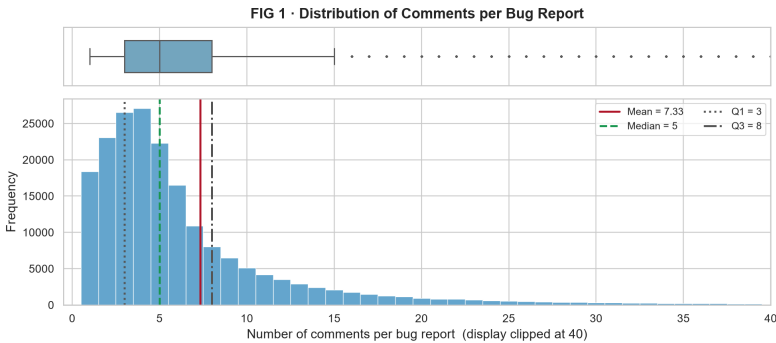


TABLE 1 · Descriptive statistics — comments per bug

Statistic	Value	Statistic	Value
n (valid)	196,633	Q3 (P75)	8
Mean	7.33	P90	14
Median (Q2)	5	P95	21
Mode	4	P99	52
Std. deviation	10.49	Maximum	429
Variance	109.95	IQR (Q3–Q1)	5
Minimum	1	Skewness	7.04
Q1 (P25)	3	Kurtosis	90.86

Tukey upper fence $Q3 + 1.5 \times IQR = 15.5$; 17,183 observations (8.74%) lie above it.

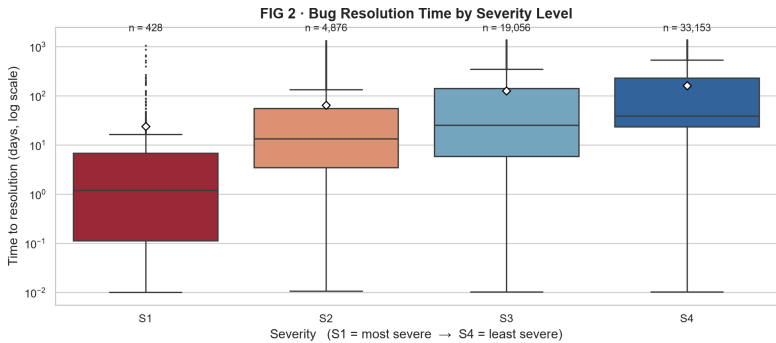


TABLE 2 · Five-number summary of resolution time (days) by severity

Severity	n	Min	Q1	Median	Q3	Max	IQR	Mean
S1	428	0.01	0.11	1.21	6.88	1,051	6.77	24.28
S2	4,876	0.01	3.53	13.54	55.87	1,323	52.35	65.12
S3	19,056	0.01	5.92	25.28	143.96	1,365	138.04	127.67
S4	33,153	0.01	23.72	39.25	231.26	1,380	207.54	164.11

Kruskal–Wallis $H = 2,765.7$, $df = 3$, $p < 0.001$ → the four medians are not all equal. 490 records resolved in < 0.01 d (≈ 15 min) excluded as automated closures.

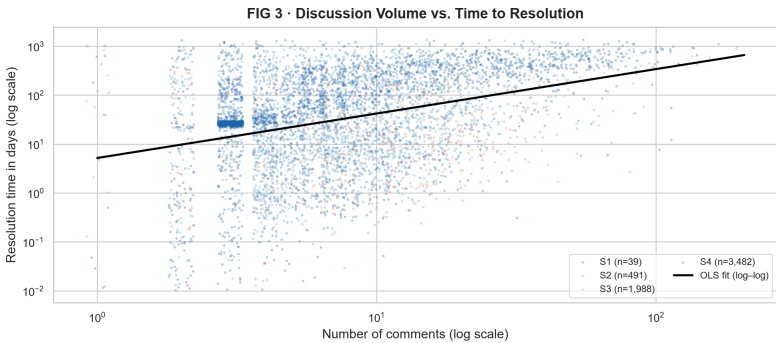


TABLE 3 · Bivariate association — comments vs. resolution time

Measure	Value	Measure	Value
n (plotted sample)	6,000	OLS slope (log–log)	0.9085
Spearman ρ	0.3175	OLS intercept	0.7172
p-value (p)	< 0.001	Std. error of slope	0.0310
Pearson r (log–log)	0.3542	Coefficient of determination r^2	0.1254

$r^2 = 0.1254$ → the log-linear model accounts for 12.5% of the variance in resolution time.

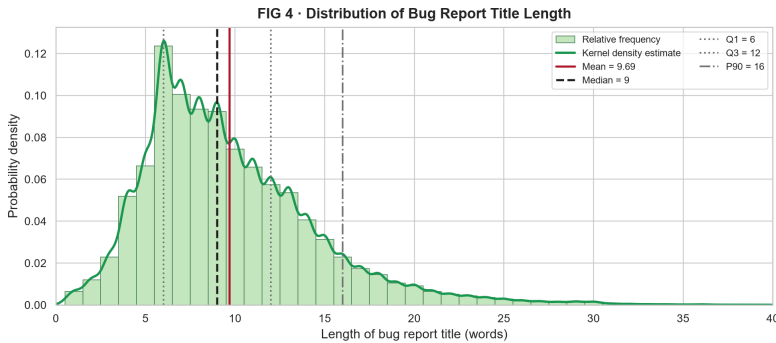


TABLE 4 · Distribution of title length (words)

Statistic	Value	Statistic	Value
n	196,633	Q3 (P75)	12
Mean	9.69	P90	16
Median	9	P99	26
Std. deviation	4.87	Maximum	57
P5	4	Skewness	1.35
Q1 (P25)	6	Kurtosis	3.19

Shapiro–Wilk on a random sample of $n = 4,000$: $W = 0.9057$, $p < 0.001$ → normality is rejected.

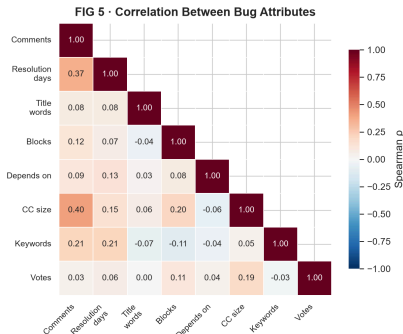


TABLE 5 · Strongest pairwise rank correlations

Rank	Variable A	Variable B	Spearman ρ	Strength
1	CC list size	Comments per bug	+0.400	moderate
2	Resolution time (days)	Comments per bug	+0.365	moderate
3	Keywords assigned	Comments per bug	+0.214	weak
4	Keywords assigned	Resolution time (days)	+0.212	weak
5	CC list size	Bugs blocked	+0.197	weak
6	Votes	CC list size	+0.187	weak
7	CC list size	Resolution time (days)	+0.150	weak
8	Dependencies	Resolution time (days)	+0.126	weak

Spearman ρ used throughout because every variable is strongly right-skewed and the relationships are monotonic rather than linear.

Dataset, Variables & Method

PROVENANCE

Source file Bug_meta_data.csv
File size 350,195,270 bytes
SHA-256 (first 24) 17457b6c451e8462c49c9867...
Columns in file 299 (20 retained)
Licence CC-BY 4.0

COMPOSITION

Type · defect 117,806
Type · task 54,844
Type · enhancement 23,983
Resolution · FIXED 83,554
Resolution · DUPLICATE 18,796
Resolution · INCOMPLETE 22,364
Resolution · INVALID 14,384
Still unresolved 45,449

SEVERITY DISTRIBUTION

S1 (most severe) 469
S2 5,615
S3 33,556
S4 (least severe) 40,202

"..." unset 101,739

STATISTICAL METHODS

- Central tendency & dispersion: mean, median, mode, variance, SD, IQR.
- Position: quartiles Q1–Q3; percentiles P5, P90, P95, P99.
- Shape: Fisher skewness, excess kurtosis. Outliers: Tukey $1.5 \times IQR$ fence.
- Association: Spearman ρ — preferred to Pearson because every variable is heavily right-skewed and the relations are monotonic, not linear.
- Group comparison: Kruskal–Wallis H (non-parametric one-way ANOVA).
- Normality: Shapiro–Wilk W on a random sub-sample ($n = 4,000$).
- Logarithmic axes where ranges span several orders of magnitude.

CLEANING DECISIONS

- De-duplicated on bug ID, keeping the first occurrence.
- Negative durations (resolved before created) set to missing.
- Resolutions under 15 min excluded from Figs 2–3 as automated closures.
- Legacy severity words excluded ($< 0.6\%$ of records; the repository had migrated to the S1–S4 scale before this window).

REFERENCE

BugsRepo: A Comprehensive Curated Dataset of Bug Reports, Comments and Contributors Information from Bugzilla. EASE 2025, Data & Tool Track. arXiv:2504.18806 · doi.org/10.5281/zenodo.15004067

FIG 6 · Where Bugs Are Reported — Product and Component Concentration

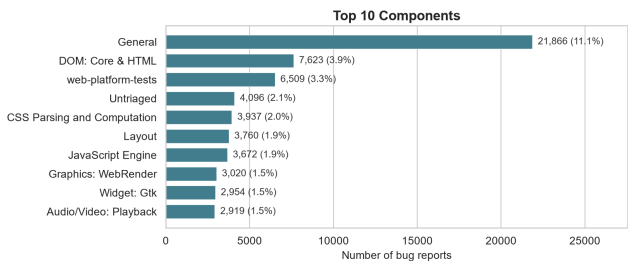
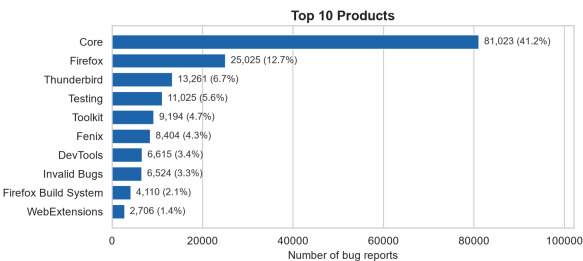


TABLE 6 · Largest categories and concentration of the label space

Rank	Product	n	% of all	Component	n	% of all
1	Core	81,023	41.2%	General	21,866	11.1%
2	Firefox	25,025	12.7%	DOM: Core & HTML	7,623	3.9%
3	Thunderbird	13,261	6.7%	web-platform-tests	6,509	3.3%
4	Testing	11,025	5.6%	Untriaged	4,096	2.1%
5	Toolkit	9,194	4.7%	CSS Parsing and Computation	3,937	2.0%

85 products and 725 components in total. Top 5 cover 71.0% of bugs by product but only 22.4% by component; 8 products vs 128 components are needed to reach 80% — the component label space is far more long-tailed. Median category size: 127 (product), 46 (component).